

Case Swarm Chat

Azure SignalR Live Chat (Option D)

Operational guide: architecture, permissions, setup, and difficulties
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Source markdown: docs/Case-Swarm-SignalR-Option-D-Guide.md. Feature flags: Teams_Bot_Config__mdt.Default Use_Azure_Chat_Bridge__c + Use_Azure_SignalR__c.

1. What Option D is

Option D replaces continuous Graph history polling as the primary live path with Azure SignalR Service (Serverless). The agent opens Case Swarm Chat; the LWC loads one-shot history, negotiates SignalR, joins the chat group, and listens for newMessage. When someone types in Teams, Bot Framework delivers type=message to the Azure relay, which pushes a DTO to SignalR group=chatId. The LWC appends immediately without a Salesforce Daily REST API hit per message. Teams remains source of truth; persistTranscript still pulls Graph into Swarm_Message__c once at end of session.

Path	Cost / risk while chat is open
Apex -> Graph ~5s	Salesforce callouts + Daily API
LWC -> Azure -> Graph ~5s	Graph 10/10s throttle; Azure executions
SignalR push (D)	Near-zero Graph while connected; SignalR + Functions on events

2. Architecture

```
Teams group chat
|- install / Adaptive Card invoke -> teamsBotRelay -> Salesforce Apex REST
`- plain type=message (RSC ChatMessage.Read.Chat OR @mention)
    -> teamsBotRelay -> Azure SignalR group=chatId target=newMessage

Salesforce LWC (caseSwarmChatCore)
|- getChatBridgeSession (HMAC + flags)
|- GET /api/chatHistory (one-shot on open; poll only if SignalR drops)
|- POST /api/negotiate -> { url, accessToken }
|- WebSocket wss://...signalr.net (native JSON hub client)
|- POST /api/joinChat -> AddToGroup(userId, chatId)
`- POST /api/chatSend -> Bot Framework + SignalR outbound echo
```

2.1 Auth surfaces (do not conflate)

Surface	Credential / resource	Purpose
Microsoft Graph	MS_Graph	Chat, members, install Teams app, list messages
Bot Framework	MS_Bot_Framework / Azure Bot	Post as bot; activities to relay
Azure SignalR	AzureSignalRConnectionString	Negotiate + group push

2.2 Hub conventions

Item	Value
Hub name	swarmChat
Group name	Teams conversation.id / Case Swarm_Team_Id__c
Client event	newMessage (one DTO argument)
SignalR user id	Salesforce UserInfo.getUserId()

3. Permissions checklist (complete)

3.1 Entra app for MS_Graph (Application + admin consent)

Permission	Why
Group.ReadWrite.All	M365 group / team flows
Team.Create / team member perms	Team swarms
Directory.Read.All	Resolve users
Chat.ReadWrite.All	Create chats, members, list messages
TeamsAppInstallation.ReadWriteAndConsentForChat.All	REQUIRED to install RSC app and consent ChatMessage.Read.Chat

TeamsAppInstallation.ReadWriteForChat.All alone is NOT enough once the Teams app declares RSC. Graph returns 403 listing the ...AndConsent... permissions.

3.2 Teams app manifest RSC (no @mention)

Without ChatMessage.Read.Chat RSC, group-chat bots only receive messages when @mentioned. Manifest v1.0.1+ must include webApplicationInfo.id = bot App Id and authorization.permissions.resourceSpecific ChatMessage.Read.Chat Application.

```
"webApplicationInfo": { "id": "<bot Microsoft App Id>", "resource": "https://Api" },
"authorization": {
  "permissions": {
    "resourceSpecific": [
      { "name": "ChatMessage.Read.Chat", "type": "Application" }
    ]
  }
}
```

3.3 Graph install body (Apex TeamsSwarmService.installBotApp)

```
POST /v1.0/chats/{chatId}/installedApps
{
  "teamsApp@odata.bind": "https://graph.microsoft.com/v1.0/appCatalogs/teamsApps/{catalogId}",
  "consentedPermissionSet": {
    "resourceSpecificPermissions": [
      { "permissionValue": "ChatMessage.Read.Chat", "permissionType": "Application" }
    ]
  }
}
```

Missing consentedPermissionSet -> ResourceSpecificPermissionsMismatch. Missing Consent Graph permission -> 403 Forbidden.

3.4 Salesforce CMDT

Setting	Purpose
Use_Azure_Chat_Bridge__c	LWC uses Azure history/send
Use_Azure_SignalR__c	negotiate/join/push (requires bridge on)
Chat_Bridge_Azure_Base_Url__c	Function App base URL with https://
Chat_Bridge_HMAC_Secret__c	Must match Azure CHAT_BRIDGE_HMAC_SECRET
Teams_App_Catalog_Id__c	Graph appCatalogs/teamsApps id (not manifest GUID)

3.5 Azure Function App settings

- AzureSignalRConnectionString - Serverless SignalR connection string
- CHAT_BRIDGE_HMAC_SECRET - must match Salesforce CMDT secret
- SALESFORCE_* - relay to Apex for Adaptive Cards only
- Existing Bot/Graph settings used by chatSend

3.6 CSP Trusted Sites (LEX connect-src)

Site	Endpoint
Chat_Bridge_Azure_Connect	https://<function-app>.azurewebsites.net
Chat_Bridge_SignalR_Connect	https://case-swarm-signalr.service.signalr.net
Chat_Bridge_SignalR_Wss	wss://case-swarm-signalr.service.signalr.net

https and wss are SEPARATE allowlist entries. Missing wss blocks the WebSocket even when https is present. Azure CSP must include the https:// scheme.

4. Setup steps (ordered)

4.1 Azure SignalR + Function App

1. Create Azure SignalR Service: Serverless (Free_F1 fine for dev).
2. Set Function App AzureSignalRConnectionString from primary connection string.
3. Deploy relay with negotiate, joinChat, SignalR outputs on teamsBotRelay/chatSend.
4. Confirm Functions extension bundle supports SignalR bindings.
5. Set CHAT_BRIDGE_HMAC_SECRET identical to Salesforce CMDT.

4.2 Teams app (RSC)

1. Update teamsapp/manifest.json with RSC + webApplicationInfo.id = bot App Id.
2. Bump version; zip manifest + icons; upload to Teams Admin catalog.
3. Copy catalog teamsApps id into Teams_App_Catalog_Id__c.
4. Entra Graph app: ReadWriteAndConsentForChat.All + Grant admin consent.

4.3 Salesforce

1. Deploy Apex/LWC/CMDT/CSP including Chat_Bridge_SignalR_Wss.
2. Set CMDT: base URL, HMAC, both feature flags true, catalog id.
3. Confirm installBotApp sends consentedPermissionSet.

4.4 Smoke test

1. Start a NEW Chat swarm (old chats may lack RSC install).
2. Bot install succeeds (Swarm_Bot_Install_Error__c empty).
3. Case chat badge: Live: SignalR push.
4. Network: WebSocket to *.service.signalr.net; negotiate/joinChat 200.
5. Type in Teams WITHOUT @mention -> LWC updates instantly.
6. Optional: Sync / ~30s confirms Graph backup poll.

5. Client implementation notes

5.1 Native WebSocket hub (not @microsoft/signalr static resource)

Loading @microsoft/signalr via loadScript under Lightning Web Security failed repeatedly (see section 6). Production uses caseSwarmChatSignalR.js: minimal JSON hub protocol, handshake, newMessage invocations, ping/pong, reconnect with fresh negotiate.

5.2 Polling while SignalR is connected

During bring-up (before RSC), the LWC kept a ~30s Graph chatHistory poll even while SignalR showed connected. That backup is OFF now: while SignalR is connected there is no continuous poll; poll resumes only on disconnect fallback.

5.3 Session / HMAC

- Apex mints short-lived HMAC for chatId + userId.
- Negotiate requires x-signalr-userid matching token userId.

- joinChat adds that user to SignalR group chatId. *Case Swarm - Azure SignalR Live Chat (Option D)*

6. Difficulties faced (and fixes)

Real blockers encountered bringing Option D live in this project:

6.1 CSP Azure URL missing https://

Symptom: fell back to ~5s poll; negotiate blocked. Cause: Chat_Bridge_Azure_Connect host without scheme. Fix: full https:// Function App URL.

6.2 require is not defined (SignalR static resource)

Symptom: badge showed require is not defined. Cause: LWS injects CommonJS module/exports; UMD took Node path. Attempted UMD patches; insufficient.

6.3 require(abort-controller) / AbortController unavailable

Symptom: LWS hides free AbortController; SignalR Node polyfill path. Attempted polyfill + fetch bridge; still fragile under LWS.

6.4 loadLibrary timed out after 20000ms

Symptom: ~5ms download but loadScript never settled. Cause: LWS evaluation hang (unreliable with bare self). Fix: abandoned loadScript/@microsoft/signalr; native WebSocket client in LWC.

6.5 CSP blocks wss:// despite https:// allowlist

Symptom: CSP violation on WebSocket. Cause: Salesforce lists https and wss separately. Fix: Chat_Bridge_SignalR_Wss = wss://case-swarm-signalr.service.signalr.net.

6.6 Connected but no Teams messages unless @mention

Symptom: green badge; plain text never arrives; @mention works. Cause: group bots only get mentions without RSC. Fix: manifest RSC + install consentedPermissionSet + Consent Graph permission. Interim during bring-up: ~30s Graph backup poll (later removed).

6.7 ResourceSpecificPermissionsMismatch on bot install

Symptom: install fails after adding RSC to manifest. Cause: POST lacked consentedPermissionSet. Fix: Apex installBotApp includes ChatMessage.Read.Chat Application consent.

6.8 403 missing ...ReadWriteAndConsentForChat.All

Symptom: after body fix, Graph 403 listing Consent permissions. Cause: Entra app only had ReadWriteForChat.All. Fix: add ReadWriteAndConsentForChat.All, admin consent, new swarm.

6.9 Public static resource CDN stickiness

Symptom: patched signalr JS not used after deploy. Cause: cacheControl=Public. Lesson: Private or new resource name; moot after native client.

6.10 Concurrent loadScript from multiple LWC hosts

Symptom: bare undefined when widget + record + utility all mounted. Lesson: shared module promise if using loadScript; native client avoids this.

7. Runtime behaviour reference

Badge	Meaning
Live: SignalR push	WebSocket up; push only (no continuous Graph poll)
SignalR: ...	Connecting / error / poll fallback

Badge	Meaning
Live: Azure poll (~5s)	Bridge on, SignalR off
Live: Apex poll (~5s)	Bridge off

Inbound path	When it works
SignalR push	Bot received activity (RSC or @mention) -> relay -> group
Graph poll fallback	Only when SignalR is down / reconnecting
Outbound SF -> Teams	chatSend -> Bot + SignalR echo

8. Troubleshooting quick matrix

Symptom	Check
Still polling every 5s	Flags, CSP https, badge error text
CSP / wss blocked	Chat_Bridge_SignalR_Wss present
negotiate 401	HMAC secret match; x-signalr-userid
joinChat 401	Token chatId match
Push only on @mention	RSC uploaded? Consent install? Graph Consent perm?
Install PermissionsMismatch	consentedPermissionSet? Catalog = RSC app?
Install 403 Consent	Entra ...AndConsent...All + admin consent
Cards work, push doesn't	Relay SignalR output? Connection string set?

9. Key repo files

Area	Path
LWC core	force-app/.../lwc/caseSwarmChatCore/
Native SignalR client	.../caseSwarmChatSignalR.js
Apex session / bridge	TeamsSwarmChatController.cls
Bot install + RSC body	TeamsSwarmService.installBotApp
Relay push	relay/src/functions/teamsBotRelay.js
Negotiate / join	relay/src/functions/negotiate.js, joinChat.js
Hub helpers	relay/src/signalrHub.js
Teams manifest	teamsapp/manifest.json
CSP	cspTrustedSites/Chat_Bridge_*
SignalR Azure notes	relay/docs/signalr-setup.md

10. Rollback

1. Use `_Azure_SignalR__c = false` -> Azure ~5s poll (bridge still on).
2. Use `_Azure_Chat_Bridge__c = false` -> Apex Graph poll.
3. Optional: remove RSC from Teams manifest and revert install body if Consent permission must be dropped (returns @mention requirement for bot-received messages).

11. Summary

Option D is viable in Salesforce LWC if:

1. You avoid shipping `@microsoft/signalr` through `loadScript` under LWS (use a native WebSocket hub client).
2. You allowlist both `https` and `wss` for SignalR.
3. You treat RSC + Consent Graph permission + `consentedPermissionSet` as mandatory for no-@mention push.

4. Optionally keep a slow Graph backup during RSC bring-up; once push is proven, stop polling while SignalR is connected.

Once those are in place: green Live: SignalR push, Teams text without @mention, Adaptive Cards unchanged via relay -> Salesforce.